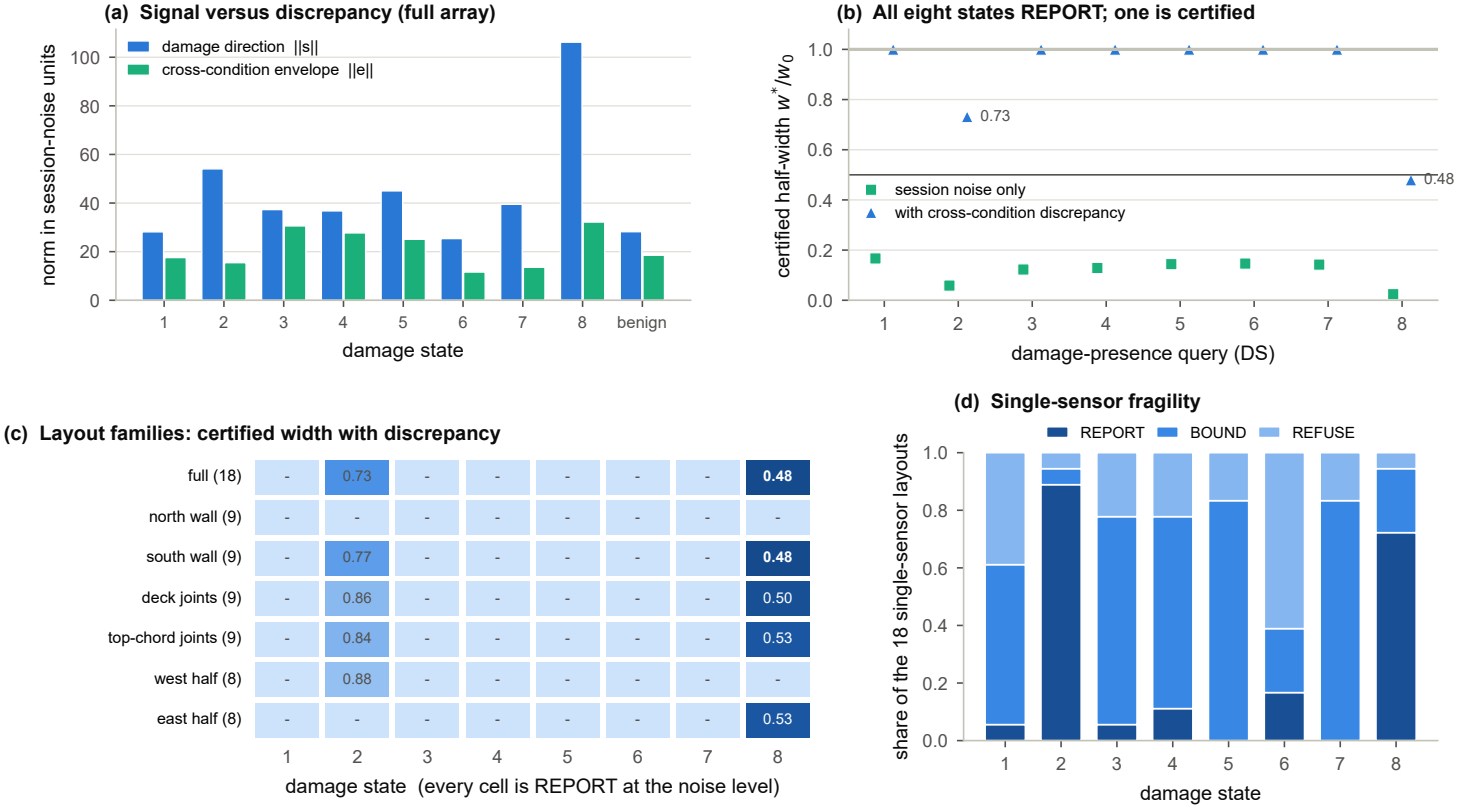

(a) Signal versus discrepancy (full array)
damage direction ||s||
cross-condition envelope ||e||
norm in session-noise units
damage state
(b) All eight states REPORT; one is certified
certified half-width w*/w0
session noise only
with cross-condition discrepancy
0.73
0.48
damage-presence query (DS)
(c) Layout families: certified width with discrepancy
full (18)
north wall (9)
south wall (9)
deck joints (9)
top-chord joints (9)
west half (8)
east half (8)
damage state (every cell is REPORT at the noise level)
(d) Single-sensor fragility
REPORT
BOUND
REFUSE
share of the 18 single-sensor layouts
damage state